

1. One-sided limits

Let f be defined on an interval immediately to the right of a . The statement

$$\lim_{x \rightarrow a^+} f(x) = L$$

means that for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$0 < x - a < \delta \implies |f(x) - L| < \varepsilon.$$

The implication must hold for every x in the domain satisfying its hypothesis. Similarly, the left-hand limit $\lim_{x \rightarrow a^-} f(x) = L$ uses

$$0 < a - x < \delta \implies |f(x) - L| < \varepsilon.$$

The value $f(a)$ is not used in either definition.

Theorem. Suppose f is defined on both sides of a sufficiently near it. Then

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x).$$

Proof. If the two-sided limit is L , the same δ works on each side. Conversely, let $\varepsilon > 0$. Choose $\delta_-, \delta_+ > 0$ from the left- and right-hand definitions, and set $\delta = \min\{\delta_-, \delta_+\}$. For $0 < |x - a| < \delta$, either $x < a$ or $x > a$. The corresponding one-sided inequality gives $|f(x) - L| < \varepsilon$.

Uniqueness. One-sided limits are unique. For example, if the right-hand limits were distinct numbers L, M , apply both definitions with $\varepsilon = |L - M|/3$. Choose $x > a$ close enough that both estimates hold. Then

$$|L - M| \leq |L - f(x)| + |f(x) - M| < \frac{2}{3}|L - M|,$$

a contradiction. The left-hand proof is the same with $x < a$. Thus different left- and right-hand limits rule out a two-sided limit.

Example

For $f(x) = |x|/x$,

$$\lim_{x \rightarrow 0^-} f(x) = -1, \quad \lim_{x \rightarrow 0^+} f(x) = 1.$$

Each assertion follows from a constant value on the relevant side. The criterion gives another proof that the two-sided limit does not exist.

Limit laws on one side. The algebraic limit laws also hold for one-sided limits, with the same nonzero-denominator condition. Their proofs use the same estimates and minima of distances, restricting every x to the chosen side.

2. Continuity

Let $f : D \rightarrow \mathbb{R}$ and $a \in D$. **Definition.** The function f is *continuous at a* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$x \in D, \quad |x - a| < \delta \quad \implies \quad |f(x) - f(a)| < \varepsilon.$$

Here the comparison value is $f(a)$, and $x = a$ is allowed. At that point the difference is zero, so the inequality holds automatically.

If a is an interior point of an interval in the domain, this is equivalent to

$$\boxed{\lim_{x \rightarrow a} f(x) = f(a).}$$

Indeed, excluding $x = a$ gives the limit definition, and including $x = a$ adds only the inequality $0 < \varepsilon$.

Endpoints. On a closed interval $[u, v]$, continuity at u means $\lim_{x \rightarrow u^+} f(x) = f(u)$, and continuity at v means $\lim_{x \rightarrow v^-} f(x) = f(v)$. All approaches remain within the domain. A function is continuous on an interval if it is continuous at every point of that interval, with this convention at included endpoints.

Example

For a real number k , define

$$F_k(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x \neq 1, \\ k, & x = 1. \end{cases}$$

For $x \neq 1$, the quotient is $x + 1$, so $\lim_{x \rightarrow 1} F_k(x) = 2$, independently of k . Thus F_k is continuous at 1 exactly when $k = 2$. For this value, a direct proof takes $\delta = \varepsilon$, since

$$|F_2(x) - F_2(1)| = |x - 1|$$

for every x , including $x = 1$.

The original quotient, with domain $\mathbb{R} \setminus \{1\}$, is not defined at 1. Assigning the value 2 gives a continuous extension. Assigning any other value leaves a discontinuity there, although the limit still exists.

3. Arithmetic operations

Theorem. Suppose f and g are continuous at a on a common interval, and $c \in \mathbb{R}$. Then

$$f + g, \quad cf, \quad f - g, \quad fg$$

are continuous at a . If also $g(a) \neq 0$, then f/g is defined sufficiently near a within the interval and is continuous at a .

Proof. At an interior point, continuity gives $f(x) \rightarrow f(a)$ and $g(x) \rightarrow g(a)$ as $x \rightarrow a$. The sum and scalar-multiple limit laws imply

$$\begin{aligned} \lim_{x \rightarrow a} (f(x) + g(x)) &= f(a) + g(a) = (f + g)(a), \\ \lim_{x \rightarrow a} cf(x) &= cf(a) = (cf)(a). \end{aligned}$$

Both limits equal the corresponding function values. Thus the sum and scalar multiple are continuous. Since $f - g = f + (-1)g$, the difference is continuous as well.

The product law gives

$$\lim_{x \rightarrow a} f(x)g(x) = f(a)g(a) = (fg)(a),$$

which proves continuity of the product.

For the quotient, first apply continuity of g with $\varepsilon = |g(a)|/2 > 0$. There is an $r > 0$ such that

$$|x - a| < r \implies |g(x)| \geq |g(a)| - |g(x) - g(a)| > \frac{|g(a)|}{2} > 0.$$

The quotient is therefore defined on this neighborhood. The quotient limit law now gives

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{f(a)}{g(a)} = (f/g)(a).$$

This proves continuity. At an included endpoint, use the corresponding one-sided limits; the same arguments apply.

Consequences. The identity function and constant functions are continuous. Repeated products, scalar multiples and finite sums show that every polynomial is continuous on $\mathbb{R}$. A rational function P/Q is continuous at every point where $Q \neq 0$.

Example

The function

$$R(x) = \frac{x^2 + 1}{x^2 - 4}$$

is continuous on $(-\infty, -2)$, $(-2, 2)$ and $(2, \infty)$. It is not defined at ± 2 . The quotient theorem applies at every point of its domain.

4. Composition

Theorem. Let $f : D \rightarrow E$ and $g : E \rightarrow \mathbb{R}$, with $a \in D$. If f is continuous at a and g is continuous at $b = f(a)$, then $g \circ f$ is continuous at a .

Proof. Let $\varepsilon > 0$. Continuity of g at b gives a number $\eta > 0$ such that

$$y \in E, \quad |y - b| < \eta \implies |g(y) - g(b)| < \varepsilon.$$

Now apply continuity of f at a , using the positive number η . There exists $\delta > 0$ such that

$$x \in D, \quad |x - a| < \delta \implies |f(x) - f(a)| < \eta.$$

For such x , take $y = f(x) \in E$. Since $b = f(a)$, the first implication gives

$$|g(f(x)) - g(f(a))| < \varepsilon.$$

This is continuity of $g \circ f$ at a .

The order of the choices is

$$\varepsilon \xrightarrow{\text{continuity of } g \text{ at } f(a)} \eta \xrightarrow{\text{continuity of } f \text{ at } a} \delta.$$

The first implication permits $y = b$. Consequently the proof still works when $f(x) = f(a)$ at points other than a .

Example

Take $f(x) = x^2$ and $g(y) = 1/(y - 3)$. At $a = 2$, the inner function is continuous and has value $f(2) = 4$. The outer function is continuous at 4, since $4 - 3 \neq 0$. Therefore $1/(x^2 - 3)$ is continuous at 2, and its limit there is 1. For the theorem, restrict the inner domain to $D = (\sqrt{3}, \infty)$ and take $E = (3, \infty)$.

A related limit statement. If $f(x) \rightarrow b$ as $x \rightarrow a$, g is continuous at b , and $f(x)$ lies in the domain of g near a , then

$$g(f(x)) \rightarrow g(b).$$

The proof is the same, with $0 < |x - a| < \delta$ in the inner implication. No value $f(a)$ is needed for this limit statement.

5. Roots

Theorem. The function $x \mapsto \sqrt{x}$ is continuous on $[0, \infty)$, with right-hand continuity at zero.

Proof at $a > 0$. For $x \geq 0$, rationalization gives

$$|\sqrt{x} - \sqrt{a}| = \frac{|x - a|}{\sqrt{x} + \sqrt{a}} \leq \frac{|x - a|}{\sqrt{a}}.$$

Given $\varepsilon > 0$, take $\delta = \varepsilon\sqrt{a} > 0$. For every domain point with $|x - a| < \delta$, the displayed bound is less than ε .

Proof at zero. Given $\varepsilon > 0$, take $\delta = \varepsilon^2$. If $x \geq 0$ and $|x| < \delta$, then

$$|\sqrt{x} - \sqrt{0}| = \sqrt{x} < \sqrt{\delta} = \varepsilon.$$

This also covers $x = 0$. Thus the endpoint is included in the theorem.

Other roots. For each integer $n \geq 2$, the nonnegative n th root is continuous on $[0, \infty)$. At $a > 0$, put $u = x^{1/n}$ and $b = a^{1/n} > 0$. Factorization gives

$$x - a = (u - b)(u^{n-1} + u^{n-2}b + \cdots + b^{n-1}).$$

Every term in the second factor is nonnegative and the last is $b^{n-1} > 0$. Hence

$$|x^{1/n} - a^{1/n}| \leq \frac{|x - a|}{b^{n-1}}.$$

Take $\delta = \varepsilon b^{n-1}$. At zero, use $\delta = \varepsilon^n$.

For odd n , the real n th root is also defined for negative numbers, with $x^{1/n} = -(-x)^{1/n}$ for $x < 0$. Continuity on $(-\infty, 0)$ follows from composition and scalar multiplication applied to the positive-root result. At zero, the identity $|x^{1/n}| = |x|^{1/n}$ gives the same choice $\delta = \varepsilon^n$, now from both sides.

Example

The function $\sqrt{1 + x^2}$ is continuous on $\mathbb{R}$: the inner polynomial is continuous and takes values in $[1, \infty)$, where the square-root function is continuous. In particular,

$$\lim_{x \rightarrow 2} \sqrt{1 + x^2} = \sqrt{5}.$$

6. The Intermediate Value Theorem

Theorem. Let f be continuous on $[a, b]$, with $a < b$. If N lies strictly between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ such that

$$f(c) = N.$$

If equality with an endpoint value is allowed, the corresponding conclusion is $c \in [a, b]$.

In particular, if $f(a)f(b) < 0$, then f has a zero in (a, b) . This gives existence; uniqueness requires a separate argument.

Example

Show that the equation $x^3 + x = 1$ has a solution in $(0, 1)$. Set $F(x) = x^3 + x - 1$. This polynomial is continuous on $[0, 1]$, and

$$F(0) = -1 < 0, \quad F(1) = 1 > 0.$$

The theorem gives $c \in (0, 1)$ with $F(c) = 0$. For a smaller interval, exact arithmetic gives

$$F(1/2) = -\frac{3}{8} < 0, \quad F(3/4) = \frac{11}{64} > 0.$$

Thus there is a solution in $(1/2, 3/4)$.

To prove uniqueness, let $y > x$. Then

$$F(y) - F(x) = (y - x)(y^2 + xy + x^2 + 1) > 0,$$

because $y^2 + xy + x^2 + 1 = (y + x/2)^2 + 3x^2/4 + 1 > 0$. So F is strictly increasing and cannot have two zeros. This uniqueness argument uses algebra, without derivatives.

The continuity hypothesis. Define h on $[-1, 1]$ by

$$h(x) = \begin{cases} -1, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

The endpoint values have opposite signs, but h never equals zero. The theorem does not apply because h is discontinuous at zero.

Uniqueness is not part of the conclusion. The continuous polynomial $x^3 - x$ has opposite signs at -2 and 2 , and has three zeros: $-1, 0, 1$.

Reading. Stewart, *Calculus: Early Transcendentals*, 9th ed., §2.2 (one-sided limits) and §2.5 (continuity and the Intermediate Value Theorem).